

Panel Methods & DID I

Econometrics for Finance

Morteza Aghajanzadeh

Stockholm School of Economics

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Identification Arsenal

Recap

Our arsenal

- Use **Matching** to create balance on observables between the treatment and control groups.
- Use **Instrumental Variable (IV)** to isolate exogenous variation in the treatment.
- Use **Panel Data** or **Repeated Cross-section** to control for unobserved heterogeneity:
 - ▶ **Difference-in-Differences (DiD)**
 - ▶ **Synthetic Controls (SC)**
 - ▶ **Event study**
- Use **Regression Discontinuity Design (RDD)** to exploit exogenous variation in the treatment around a cutoff.

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- The 2×2 DiD

Panel Data

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 - 1 i : N individuals
 - 2 t : T periods
- What is the difference between panel data and time-series data, or repeated cross-section data?
- Panel data can be **balanced** or **unbalanced**. We have to be careful about this.
 - ▶ If the panel is unbalanced, missing observations should be random, i.e., unrelated to the error term μ_{it} .

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- **More Data Points:** With both cross-sectional and time-series dimensions, panel data provide more data points, increasing degrees of freedom and reducing collinearity among explanatory variables.
- **Identifying Effects:** Panel data enable the identification and measurement of effects that are simply not detectable in pure cross-section or time-series data.

Example of Panel Data

Wide Format

Team	Points	Assists	Rebounds
A	88	12	22
B	91	17	28
C	99	24	30
D	94	28	31

Long Format

Team	Variable	Value
A	Points	88
A	Assists	12
A	Rebounds	22
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B	Assists	17
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- What do you think about a_i ?
- Let's keep it simple and assume that the first case is true.

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(Example: the number of individual training hours is not correlated with the ability of the individual)
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This is the first assumption we need to make with panel data, but we still have to be careful.

Independent individual effect

$$\text{Cov}(a_i, x_{it}) = 0$$

If a_i is random and independent of x_{it} , then we do not have the OVB problem.

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We have a serial correlation problem: for the same individual in two different periods $t \neq s$,

$$\mathbb{E}[\varepsilon_{it}\varepsilon_{is}] = \mathbb{E}[(a_i + \mu_{it})(a_i + \mu_{is})] = \mathbb{E}[a_i^2] = \sigma_a^2 \neq 0, \quad \text{Var}(\varepsilon_{it}) = \sigma_a^2 + \sigma_\mu^2$$

$$\mathbb{E}[\varepsilon\varepsilon'] = \Omega = \begin{bmatrix} \sigma_a^2 + \sigma_\mu^2 & \cdots & \sigma_a^2 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \sigma_a^2 & \cdots & \sigma_a^2 + \sigma_\mu^2 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & \sigma_a^2 + \sigma_\mu^2 & \cdots & \sigma_a^2 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \sigma_a^2 & \cdots & \sigma_a^2 + \sigma_\mu^2 \end{bmatrix}$$

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- ▶ Simply ignore the panel structure and run OLS on the pooled data.
- ▶ We have to be careful about the standard errors.
- ▶ We need to cluster the standard errors at the individual level.

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Note: Recall the main assumption

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Is there any other solution? **Hint:** What additional features does panel data provide that are not available in cross-sectional data?

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- We can control for a_i . This is the **Fixed Effects (FE) model**.

Fixed Effects model

Assumption

$$y_{it} = \alpha + \beta x_{it} + a_i + \mu_{it}$$

- We need to assume that the regressors in every period are uncorrelated with the error term in every period (**strict exogeneity**)¹:

$$\mathbb{E}(x_{is}\mu_{it}) = 0, \quad \forall s, t \in \{1, \dots, T\}$$

¹If it does not hold (e.g., with a lagged dependent variable), you can use the **Arellano–Bond** estimator.

Fixed Effects model

Estimation

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- ② **Within estimator:** Subtract each individual's time average from each variable and then run OLS on the transformed data.
- ③ **First-difference estimator:** Take the first difference of each variable and then run OLS on the transformed data. (I do not cover this one.)

Direct estimator

We add a dummy variable for each individual and run OLS on this augmented regression.

This helps to understand the intuition behind the FE model.

Write the model as:

$$y_{it} = \alpha + \beta x_{it} + \sum_{j=2}^N a_j d_{j,i} + \mu_{it}$$

- $d_{j,i}$ is a dummy variable for individual j .
- $d_{j,i} = 1$ if observation i belongs to individual j and $d_{j,i} = 0$ otherwise.
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With a large number of individuals, we have many dummy variables, which makes estimation costly.

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$$(y_{it} - \bar{y}_i) = \beta(x_{it} - \bar{x}_i) + (\mu_{it} - \bar{\mu}_i)$$

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The within estimator gives numerically the same $\hat{\beta}$ as the direct estimator (FWL theorem), but it remains feasible with a large number of individuals. (If you demean manually, the standard errors need a degrees-of-freedom correction.)

Choosing between Fixed Effects (FE) and Random Effects (RE)

- **Fixed Effects (FE):**

- ▶ Uses only within-unit variation over time.
- ▶ Removes all time-invariant unobserved heterogeneity (but not time-varying confounders).
- ▶ Good when you believe differences across units (e.g., individuals, firms) might be correlated with the independent variables.

- **Random Effects (RE):**

- ▶ Treats the individual effect a_i as random.
- ▶ Suitable when you believe the variability across units is uncorrelated with the independent variables.
- ▶ More efficient than FE, but inconsistent if this assumption fails.

- **Hausman Test:**

- ▶ A statistical test to decide between FE and RE.
- ▶ Null hypothesis: The difference in coefficients is not systematic.
- ▶ If significant, prefer FE over RE.

Practical issue

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Definition: b_t is the **time effect** or **time heterogeneity**.

- $a_i = a_1 D_{i=1} + \cdots + a_{N-1} D_{i=N-1}$
- $b_t = b_1 D_{t=1} + \cdots + b_{T-1} D_{t=T-1}$

Coefficients intuition

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- Think of running a separate regression for each individual and then taking a weighted average of the slopes, with weights proportional to the within-individual variance of x_{it} . (This is just an intuition, not exactly what is going on.)
- We only get the average effect of x_{it} on y_{it} .

Understanding the Heterogeneity

A puzzle

Example: Income and Access to Clean Water

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FE absorbs all the differences between richer and poorer households, so it may find no effect.

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- So FE may find no effect because:
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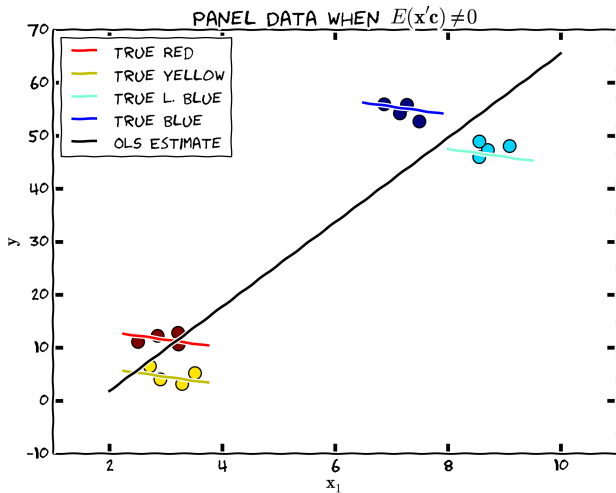
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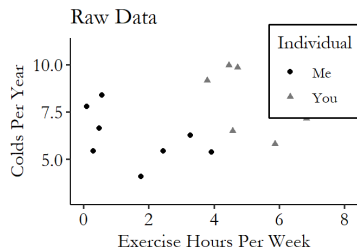
Warning

Sometimes the effect you care about lives in the cross-sectional dimension, and by using the FE model, you may lose it. Be careful about the underlying mechanisms (and remember that cross-sectional correlations can also be confounded).

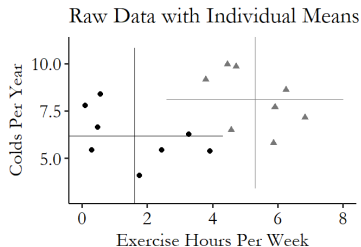
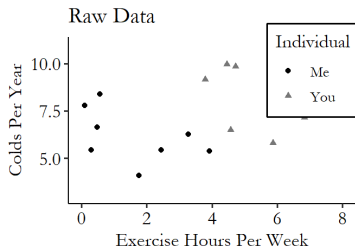
Intuition



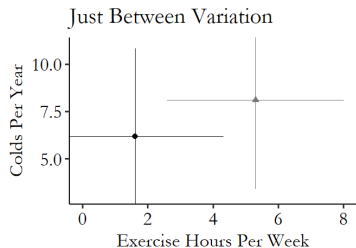
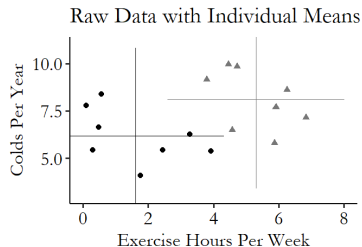
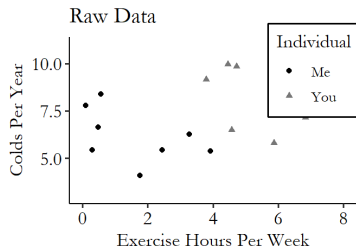
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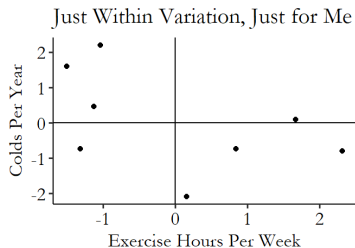
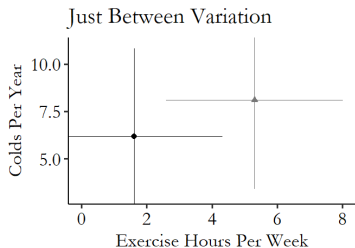
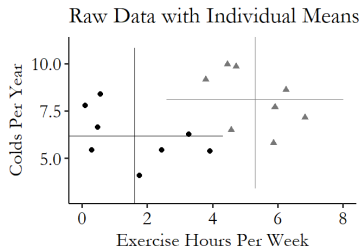
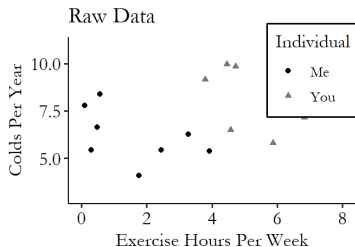
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Assignment I

- 1 The dataset “local_returns.dta” contains panel information on the monthly stock returns of firms (variable: `ret`), returns of firms in the same area (variable: `city_returns`), returns of firms in the same industry (variable: `indret`) and a few other risk factors (MRP, HML, MOM) from 2000 to 2010. The variable `permno` is the firm identifier, while `date` is the time variable. Individual firms (i.e., `permnos`) are allocated to 20 cities and 12 industries.
 - a Is it a balanced or unbalanced panel? Provide aggregate summary statistics.
 - b Report the average return and volatility of firms in the 20 cities (i.e., summary statistics by city). Do you see any differential returns among cities?
 - c We are interested in the effect of local returns on firm returns. Our model is:

$$\text{ret}_{it} = \alpha + \beta_1 \text{city_returns}_{it} + \beta_2 \text{indret}_{it} + \varepsilon_{it}$$

Estimate the model using pooled OLS and interpret the results. Test for heteroskedasticity in the residuals.

- d Which dimension should be used as a fixed effect? Estimate the model using your proposed fixed effects and interpret the results.
- e Let's try two-way fixed effects. Estimate the model using two-way (firm and time) fixed effects and interpret the results.

Assignment II

- Re-estimate the two-way fixed effects model with standard errors double-clustered by firm and time. Compare your results with the other estimations.

Extra Assignment

A Passage to America: University Funding and International Students

- 1 What is the research question?
- 2 What is the identification strategy?
- 3 What is the main result?
- 4 Try to replicate Table 3 (OLS estimates only; the rest of the table is optional).

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Motivation

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- What is panel data?
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- What if there is no panel data, or treatment is group-level?
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- Build a group-by-period panel from repeated cross-sections.
 - ▶ This controls for time-invariant group-level heterogeneity.

DiD vs. FE

$$y_{it} = a_i + b_t + \beta D_{it} + \varepsilon_{it}$$

Assume that we have a binary treatment variable D_{it} , which is equal to 1 if individual i is treated at time t . Potential outcomes from the FE point of view:

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The basic strategy is the same as FE, but we don't need repeated observations of each individual. We can use repeated cross-section data to build a group-by-period panel.

Difference-in-Differences

Simple intuition

Compare the change in average outcomes of treated vs. control groups:

- **Trend**: broad economic changes (common to both groups)
- **Effect**: treatment effect

	Before	After	After – Before
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Treated – Control:			Effect

Key assumption (parallel trends): without treatment, T changes like C .

Example

